Arithmetic Structure of the Standard Model Mass Spectrum and the Bost–Connes System

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Abstract

The masses of the 12 Standard Model fermions span 12 orders of magnitude and have previously been treated as independent experimental inputs. Starting from the Bost–Connes system, we show that the CRT decomposition of the character group $(\mathbb{Z}/7\mathbb{Z})^*$ uniquely determines six fermion sectors and the three-generation structure, and derive a unified mass formula that reduces the 12 masses to values of four Dirichlet L-functions on the critical line $s = 1/2$. The deviations for all 9 charged fermions are $\leq 2\%$. The absolute neutrino masses $m_1 = 5.113$, $m_2 = 10.01$, $m_3 = 50.67\,\text{meV}$ and mass ratio $R = 34.31$ constitute new testable predictions; m_1 and m_2 have not yet been directly measured.

Contents

1 Introduction

The fermionic mass spectrum of the Standard Model has a remarkably structured pattern. The masses of the 12 fermions—three generations of charged leptons (e, μ, τ), three generations of neutrinos (ν_1, ν_2, ν_3), and six quarks (u, d, c, s, t, b)—span 12 orders of magnitude, from the top quark at 173 GeV to neutrinos at ∼0.05 eV. The quantum numbers within each generation are identical; the only difference is mass. Intergenerational mixing is described by the CKM matrix [19] (quarks) and the PMNS matrix (neutrinos), with the Cabibbo angle $\lambda \approx 0.225$ [18] and the CP-violating phase δ_{CP} being key parameters. These mass and mixing parameters—at least 19 free parameters in total—are inputs from experimental measurements in the Standard Model; the theory itself does not predict their values [6, 21].

Yet these parameters are not independent. The lepton masses satisfy the Koide formula [3] ($Q = 2/3$ to 0.03% accuracy); the quark masses exhibit the clear Georgi–Jarlskog relation [4] ($m_b/m_\tau \approx 3$); the CKM matrix elements decrease as successive powers of λ (the Wolfenstein hierarchy [5]). These empirical regularities suggest that a unified algebraic structure underlies the mass spectrum.

We show that this structure can be found precisely within the Bost–Connes (BC) system [1, 2, 12, 31, 32, 35]. The BC system

$$\mathcal{A} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*$$

is a quantum statistical mechanical system whose automorphism group $\widehat{\mathbb{Z}}^*$ encodes arithmetic symmetries at every prime. At the prime $p = 7$, the local factor

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

carries, through its CRT decomposition, both the weak-force direction ($\mathbb{Z}/2\mathbb{Z}$) and the three-generation cycle ($\mathbb{Z}/3\mathbb{Z}$). We establish:

- The discrete quantum numbers of fermions (three generations, color, charge, spin) are precisely described by the character structure and CRT decomposition of $(\mathbb{Z}/7\mathbb{Z})^*$ (§3);
- The CKM mixing parameters are given by the phase of the Jacobi sum and the three-cycle of the Hecke action, with deviations $< 2.5\%$ (§4);
- The 12 fermion masses follow from the unified formula

$$\boxed{m_f = v \times |L|^{\text{pow}} \times \text{node factor} \times \text{Gen}} \tag{1}$$

  where L denotes a Dirichlet L-function evaluated on the critical line $s = 1/2$, with deviations $\leq 2\%$ for all 9 charged fermions (§5–6);
- The absolute neutrino masses $m_1 = 5.113$, $m_2 = 10.01$, $m_3 = 50.67$ meV are testable new predictions (§6.5).

2 Characters and Particle Classification

At the prime $p = 7$, the multiplicative group

$$(\mathbb{Z}/7\mathbb{Z})^* = \{1, 2, 3, 4, 5, 6\}$$

is a cyclic group of order six. Taking 3 as a primitive root,

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}.$$

By the Chinese Remainder Theorem,

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

This is the starting point of the paper: a minimal finite group that simultaneously carries a parity-type binary splitting and a generation-type ternary splitting.

The dual group $\widehat{(\mathbb{Z}/7\mathbb{Z})}^*$ likewise has six characters, denoted χ_n, $n = 0, 1, \dots, 5$, satisfying

$$\chi_n(3^k) = e^{2\pi i n k/6}.$$

The orders of the characters are, respectively,

$$1, 6, 3, 2, 3, 6.$$

- $(\mathbb{Z}/7\mathbb{Z})^*$ is a cyclic group of order six; the CRT decomposition endows it with two directions: $\mathbb{Z}/2\mathbb{Z}$ distinguishes symmetry type (real vs. complex), while $\mathbb{Z}/3\mathbb{Z}$ encodes the cyclic structure of internal degrees of freedom.
- The order of a character measures how deeply that sector couples to the multiplicative group—higher order detects more internal structure.
- The order distribution $\{1, 6, 3, 2, 3, 6\}$ naturally falls into four complexity levels, with the physical properties of each level determined by the order itself.

2.1 Complexity Levels

We interpret "order" as a complexity level: higher order means a longer multiplicative cycle is needed to return to the identity, corresponding to a more complex, more strongly coupled sector. This yields the following hierarchy:

Character order	Complexity	Particle sector	Description
1	Simplest	Photon/trivial gauge	Trivial vacuum representation
2	Real binary	Neutrino	Self-conjugate, Majorana tendency
3	Three-cycle	Lepton	Phase trisection structure
6	Maximal	Quark	Carries both $\mathbb{Z}/2$ and $\mathbb{Z}/3$

Here "photon" is not one of the six fermion types, but rather labels the massless trivial gauge sector corresponding to the trivial character; the actual six non-trivial fermionic degrees of freedom are given by the remaining character orbits and conjugate pairings. More precisely, the particle classification is not assigned by hand but is uniquely selected by the following four axioms.

2.2 Uniqueness Theorem

Definition 2.1 (Four Classification Axioms)**.** We seek to organize the six group elements or six characters into the six Standard Model fermion sectors, subject to:

(1) **Conjugation closure**: each physical sector must be closed under the involution $a \mapsto 7 - a$;

(2) **Three-generation compatibility**: the grouping of sectors must be compatible with $\mathbb{Z}/3\mathbb{Z}$ orbits;
(3) **Complexity monotonicity**: the orders of characters must correspond monotonically to physical complexity: order 2 is simpler than order 3, and order 3 is simpler than order 6;
(4) **Real/complex phase separation**: the unique non-trivial real character (Legendre character) must correspond exclusively to the self-conjugate sector and must not be mixed with the complex cubic characters.

Remark 2.2 (Why "higher order = more complex"?). Axiom (3), "higher character order $\Rightarrow$ more complex physics", is not an arbitrary assumption but a natural consequence of the KMS states and L-function structure of the BC system:

- **Order 1** (χ_0, trivial character): trivial under all KMS phase transitions; the BC state is trivial at all temperatures, hence $m = 0$ (gauge protection), physically the simplest;
- **Order 2** (χ_3, Legendre character): real character, purely imaginary Gauss sum $\Rightarrow$ $m = 0$ at tree level $\Rightarrow$ second simplest;
- **Order 3** (χ_2, χ_4): complex character, non-trivial Jacobi phase, mass determined by phase structure $\to$ intermediate complexity;
- **Order 6** (χ_1, χ_5): encodes all CRT directions; the richest KMS state structure, hence the most complex.

Thus complexity monotonicity is not an externally imposed assumption but a natural consequence of the KMS states [26] and the Euler product structure of L-functions in the BC system.

At first glance, there are $6! = 720$ ways to label six objects into six classes; but all 720 "ambiguities" are in fact completely eliminated by the four axioms above.

Theorem 2.3 (Uniqueness Theorem)**.** *Under the four axioms above, the assignment of the six characters of $(\mathbb{Z}/7\mathbb{Z})^*$ to the six Standard Model fermion sectors is unique up to relabeling isomorphism. Specifically: the unique order-2 real character must correspond to the neutrino sector; the unique conjugate pair of order-3 characters must correspond to the charged-lepton sector; the two pairs of order-6 characters correspond to the two quark sectors. Consequently, all 720 permutations reduce, under the axiom constraints, to a unique physical solution.*

Proof. We first consider the order structure. Among the six characters, there is exactly one trivial character of order 1, exactly one non-trivial real character of order 2, one conjugate pair of order-3 characters, and two pairs of order-6 characters. By the complexity monotonicity of Axiom (3), the three complexity levels—order 2, order 3, and order 6—are already uniquely fixed.

Turning to Axiom (4): the only non-trivial real character is the Legendre character, which is self-conjugate and therefore can only correspond to a sector that is its own antiparticle. Among fermions, this uniquely and naturally corresponds to a Majorana-type neutrino sector. Hence the order-2 level falls uniquely to neutrinos.

For the order-3 level, there is only one conjugate pair, whose phase closes on the cubic roots of unity, matching optimally the three-generation periodicity of $\mathbb{Z}/3\mathbb{Z}$. If assigned to quarks, the quarks would lose the maximal complexity of carrying both $\mathbb{Z}/2$ and $\mathbb{Z}/3$, violating Axiom (3). Thus order 3 uniquely corresponds to the charged-lepton sector.

The remaining two pairs of order-6 characters automatically correspond to the quark sectors. Because Axiom (1) requires conjugation closure and Axiom (2) requires compatibility with the three-generation orbits, any permutation attempting to swap "leptons/quarks" or "neutrinos/leptons" would violate one of: the order structure, self-conjugacy, or three-generation phase compatibility. Hence all admissible classifications are unique up to automorphism and physical relabeling. □

The assignment of particle types is thus uniquely determined by the character structure at $p = 7$. The following table summarizes the complete mapping:

Character	Order	Conjugate pair	Fermion sector	Determining axioms
χ_0	1	—	Gauge vacuum	—
χ_3	2	Self-conjugate	Neutrino	(3)(4)
χ_2, χ_4	3	One pair	Charged lepton	(1)(2)(3)
χ_1, χ_5	6	Two pairs	Quark (u-type/d-type)	(1)(2)(3)

3 Generations, Color, Charge, and Spin

The elementary particles of the Standard Model carry a set of discrete quantum numbers: three-generation structure ($e/\mu/\tau$ and the corresponding quarks and neutrinos), particle–antiparticle conjugation (C transformation), color charge (three colors for quarks, colorless leptons), electric charge quantization $(0, \pm 1/3, \pm 2/3, \pm 1)$, weak isospin ($T_3 = \pm 1/2$), and spin ($J = 1/2$) and Fermi/Bose statistics. These quantum numbers completely specify the identity of each fermion.

In this section we show that all of the above quantum information has a precise arithmetic counterpart in the local factor $(\mathbb{Z}/7\mathbb{Z})^*$ of the BC system: the three generations correspond to the three orbits of the involution, color corresponds to the resolution of the $\mathbb{Z}/3\mathbb{Z}$ direction in the CRT decomposition, charge corresponds to the multiplicative decomposition of characters, and spin corresponds to the half-order property of the generator.

3.1 Three-Generation Structure

Fermions are grouped by generation: first generation (e, ν_e, u, d), second generation (μ, ν_μ, c, s), third generation (τ, ν_τ, t, b). Fermions within the same generation are tightly linked through gauge interactions; the quantum numbers across generations are identical, with mass being the only difference— $m_e = 0.511\,\text{MeV}$ vs $m_\tau = 1777\,\text{MeV}$ (leptons), $m_u \approx 2\,\text{MeV}$ vs $m_t = 173\,\text{GeV}$ (quarks), spanning multiple orders of magnitude. Within each generation, particle and antiparticle are interchanged by the C transformation (charge conjugation), a $\mathbb{Z}/2\mathbb{Z}$ symmetry satisfying $C^2 = \text{id}$.

In the BC framework, particle–antiparticle conjugation corresponds to the involution

$$\iota(a) = 7 - a, \qquad \iota^2 = \text{id}$$

on $(\mathbb{Z}/7\mathbb{Z})^*$. Grouping the six elements into pairs under ι yields three doublets:

$$D_1 = \{1, 6\}, \qquad D_2 = \{2, 5\}, \qquad D_3 = \{3, 4\}.$$

The two elements $\{a,\, 7 - a\}$ in each doublet are exchanged by ι, corresponding to the pairing of particle and antiparticle. The three doublets correspond to the three generations.

Proposition 3.1 (Doublets and three generations)**.** *The set $\{D_1, D_2, D_3\}$ is the complete set of orbits of $(\mathbb{Z}/7\mathbb{Z})^*$ under the involution $\iota(a) = 7-a$. The particle–antiparticle pairing structure of the three-generation fermions can be precisely described by these three orbits.*

Although the three generations have identical quantum numbers, their masses differ dramatically. Within the particle–antiparticle pair of each generation, the two members have different "distances": the first-generation pair has the greatest difference (lightest, most stable), and the third-generation pair has the smallest difference (heaviest, fastest decay). This intergenerational mass hierarchy reflects the degree of asymmetry within each particle–antiparticle pair.

In the BC framework, this degree of asymmetry is described by the **tension** of the doublet. For a doublet $D = \{a, 7-a\}$, define

$$T(D) := |a - (7-a)| = |2a - 7|.$$

$T(D)$ measures the distance between the two members within the doublet: larger tension means greater asymmetry between the two members—the pair is more tightly bound to its internal structure, leaving less energy available for mass, and thus corresponding to a lighter generation. Direct computation gives

$$T(D_1) = |1-6| = 5, \qquad T(D_2) = |2-5| = 3, \qquad T(D_3) = |3-4| = 1.$$

The tension sequence is thus

$$\{5, 3, 1\}.$$

$T(D_1) = 5$ (maximum) corresponds to the lightest first generation; $T(D_3) = 1$ (minimum, two members nearly coincident) corresponds to the heaviest third generation. The ordering of tensions precisely reflects the physical mass hierarchy: this pattern holds simultaneously for charged leptons, quarks, and neutrinos— all three fermion types share the same generational structure, with the ordering of mass levels determined by the same set of doublets.

Furthermore, the tension sequence $5, 3, 1$ has a regular recursive structure— each time one crosses a generation, the asymmetry within the doublet decreases by a fixed step. In the BC framework, this regularity is precisely described by the arithmetic progression of tensions:

Theorem 3.2 (Tension Arithmetic Progression Theorem)**.** *For the three involution doublets of $p = 7$,*

$$D_1 = \{1, 6\},\ D_2 = \{2, 5\},\ D_3 = \{3, 4\},$$

the tension sequence satisfies

$$T(D_1), T(D_2), T(D_3) = 5, 3, 1,$$

forming an arithmetic progression with common difference 2. The common difference $2 = d_1$ is the order of the $\mathbb{Z}/2\mathbb{Z}$ factor in the CRT decomposition.

Proof. For $D_k = \{k, 7-k\}$ we have

$$T(D_k) = |k - (7-k)| = |2k - 7|.$$

Substituting $k = 1, 2, 3$ gives $5, 3, 1$. The adjacent differences are all 2, so this is an arithmetic progression with common difference 2. □

The tensions decrease with fixed step $d_1 = 2$, yet the mass hierarchy itself is far from uniform ($m_\tau/m_\mu \approx 17$, $m_\mu/m_e \approx 207$). This large variation arises from the nonlinear mapping by which tension is projected onto mass via the phase projection, which will be given in §6.

The three doublets $D_1 = \{1, 6\}$, $D_2 = \{2, 5\}$, $D_3 = \{3, 4\}$ are given by the $\mathbb{Z}/3\mathbb{Z}$ Galois orbits. Each doublet corresponds to one generation of fermions. Different fermion types within the same generation (leptons, quarks, neutrinos) are distinguished by the order of the character (§3.2), while the mass distribution across three generations within each type will be determined in §6 by the $\mathbb{Z}/3\mathbb{Z}$ phase projection.

3.2 Color

Quarks carry three color charges (red, green, blue), while leptons carry no color charge. In the BC framework, the color triality is encoded by the three positions $a_2 \in \{0, 1, 2\}$ in the $\mathbb{Z}/3\mathbb{Z}$ direction of the CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Whether a character can distinguish these three positions is determined by its order.

Definition 3.3 (Kernel and color)**.** The kernel of character χ_j is $\ker \chi_j = \{a \in (\mathbb{Z}/7\mathbb{Z})^* : \chi_j(a) = 1\}$, with $|\ker \chi_j| = 6/\mathrm{ord}(\chi_j)$. In physics, quarks participate in strong interactions (carry color charge) while leptons do not (colorless). This distinction is determined by the order of the character in the BC framework:

- ord $= 6$ (χ_1, χ_5): $|\ker| = 1$. Probes both the $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ directions, fully resolving all 6 elements. Corresponds physically to color-charged quarks, with color channel count $n_c = d_2 = 3$.
- ord $= 3$ (χ_2, χ_4): $|\ker| = 2$. Blind to the $\mathbb{Z}/2\mathbb{Z}$ direction. Corresponds physically to colorless charged leptons, $n_c = 1$.
- ord $= 2$ (χ_3): $|\ker| = 3$. Blind to the $\mathbb{Z}/3\mathbb{Z}$ direction, completely color-blind. Corresponds physically to colorless neutrinos, $n_c = 1$.

The kernel size $|\ker| = 6/\mathrm{ord}$ measures the "resolution" of a character with respect to the internal structure of $(\mathbb{Z}/7\mathbb{Z})^*$. Larger $|\ker|$ means more group elements are folded to the same phase, appearing externally as colorless. In physics, quarks carry color charge while leptons do not; in the BC framework, this distinction is determined by the order of the character: colored characters (ord $= 6$, $|\ker| = 1$) carry the full $(\mathbb{Z}/7\mathbb{Z})^*$ information, with the three color channels $d_2 = 3$ corresponding to the three cosets of $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}(\zeta_7)^+)$; colorless characters (ord ≤ 3) have insufficient phase information to distinguish conjugate pairs, yielding only one effective channel.

3.3 Electric Charge Quantization

Electric charge is quantized: leptons carry integer charges (0 or -1), and quarks carry fractional charges ($+2/3$ or $-1/3$). In the BC framework, electromagnetic interactions are modeled by the order-3 character χ_2. The charge Q measures the coupling strength of each sector to the χ_2 direction, and the coupling of colored sectors is divided equally among the $n_c = d_2 = 3$ color channels.

Using character multiplication $\chi_i \cdot \chi_j = \chi_{i+j \bmod 6}$ and $\bar{\chi}_2 = \chi_4$, the charges of each fermion type have a natural arithmetic interpretation in the BC framework:

- Charged leptons (e, μ, τ) have charge $Q = -1$. In BC, charged leptons correspond to χ_2, which is itself the EM character, with coupling strength 1.

- Neutrinos (ν) have charge $Q = 0$. In BC, neutrinos correspond to χ_3, which contains no χ_2 component and does not couple to electromagnetism.
- d-type quarks (d, s, b) have charge $Q = -1/3$. In BC, $\chi_5 = \chi_2 \cdot \chi_3$ contains a χ_2 component, but since quarks carry color charge, the electromagnetic coupling is divided equally among $d_2 = 3$ color channels, giving $Q = -1/d_2 = -1/3$.
- u-type quarks (u, c, t) have charge $Q = +2/3$. In BC, $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ contains $\bar{\chi}_2$, likewise divided among $d_2 = 3$ color channels. In the weak-isospin doublet (u, d), $T_3(u) = +1/2$, $T_3(d) = -1/2$. From $Q(d) = -1/3$ one obtains hypercharge $Y = Q - T_3 = -1/3 + 1/2 = +1/6$; the doublet shares hypercharge, so $Q(u) = T_3 + Y = 1/2 + 1/6 = +2/3$.

Theorem 3.4 (Charge Formula)**.**

$$Q(e, \mu, \tau) = -1 \qquad \textit{(pure } \chi_2\textit{)} \tag{2}$$

$$Q(\nu) = 0 \qquad \textit{(pure } \chi_3\textit{, no } \chi_2 \textit{ component)} \tag{3}$$

$$Q(d, s, b) = -\frac{1}{d_2} = -\frac{1}{3} \qquad \textit{(}\chi_2 \textit{ distributed over colors)} \tag{4}$$

$$Q(u, c, t) = T_3 + Y = +\frac{2}{3} \qquad \textit{(doublet structure)} \tag{5}$$

3.4 Discrete Spin and Fermion Statistics

Fermions carry spin 1/2 (a half-rotation flips the sign; a full rotation returns the state to its original value), and fermions and bosons obey different statistics. In the BC framework, these properties arise naturally from the doublet structure.

Theorem 3.5 (Discrete spin)**.** *The defining characteristic of spin* $1/2$ *in physics is: a half-rotation* (π) *flips the sign of the state vector, while a full rotation* (2π) *returns it to its original value. In the BC framework, this property can be described using the generator* $g = 3$ *(of order* 6*) of* $(\mathbb{Z}/7\mathbb{Z})^*$*. By Euler's criterion,*

$$g^{(p-1)/2} = 3^3 = 27 \equiv -1 \pmod 7,$$

i.e., the "half-turn" of the generator (g^3) *equals* -1*. On the antisymmetric representation of the doublet* $\{a, 7 - a\}$, -1 *acts as* $-\mathrm{id}$*:*

$$g^3|\psi\rangle = -|\psi\rangle \quad \textit{(half-turn sign flip)}, \qquad g^6|\psi\rangle = +|\psi\rangle \quad \textit{(full-turn restoration)}.$$

This precisely reproduces the physical behavior of spin $1/2$*.*

Proof. $3^3 = 27 = 4 \times 7 - 1 \equiv -1 \pmod 7$, which is a direct consequence of Euler's criterion. The involution $\iota\colon a \mapsto 7 - a$ on the doublet is precisely multiplication by -1. The antisymmetric state satisfies $\iota|\psi\rangle = -|\psi\rangle$, i.e., sign flip under half-turn; $\iota^2 = \mathrm{id}$ is the full-turn restoration. □

Theorem 3.6 (Fermi/Bose $\mathbb{Z}/2\mathbb{Z}$ classification)**.** *In physics, fermions and bosons obey different quantum statistics: fermionic states change sign under particle exchange (antisymmetric), while bosonic states do not (symmetric). In the BC framework, this classification is given by the eigenvalue decomposition of the involution* ι *on the doublet:*

- *Symmetric state* ($\iota = +1$)*:* $a + (7 - a) = 7$ *(constant, carries no dynamical content)—corresponds to Bose statistics;*

- *Antisymmetric state ($\iota = -1$): $a - (7 - a) = 2a - 7 = \pm T$ (tension, carries information)—corresponds to Fermi statistics.*

The Legendre character χ_3 provides the arithmetic criterion for this classification: $\chi_3(\iota) = +1$ (QR) corresponds to bosons; $\chi_3(\iota) = -1$ (QNR) corresponds to fermions.

Remark 3.7. Conventionally, spin 1/2 is defined via continuous rotations in SU(2) ($\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$). In the present framework, spin is given directly by the $\mathbb{Z}/2\mathbb{Z}$ subgroup $\{1, -1\}$ of $(\mathbb{Z}/7\mathbb{Z})^*$, without introducing a continuous group. The two descriptions are physically equivalent in their predictions.

Remark 3.8 (Half-integer spin and quadratic extensions). $-1 \equiv 6 \pmod 7$ is not a quadratic residue ($\sqrt{-1}$ does not exist in $\mathbb{F}_7$) [15]. Half-integer spin requires the quadratic extension $\mathbb{F}_{7^2}$, corresponding to the double cover $\mathrm{SO}(3) \to \mathrm{SU}(2)$. This provides an arithmetic analogue of the spin-statistics connection.

3.5 Summary of Fermion Quantum Numbers

At this point, combining the characters and orders from §2 with the CRT decomposition, doublets, and involution from §3, all quantum numbers of each fermion type have been determined:

sector	χ_j	order	Q	T_3	$SU(3)_c$	J	particles
u-type quark	χ_1	6	$+2/3$	$+1/2$	$\mathbf{3}$	1/2	u, c, t
d-type quark	χ_5	6	$-1/3$	$-1/2$	$\mathbf{3}$	1/2	d, s, b
charged lepton	χ_2	3	-1	$-1/2$	$\mathbf{1}$	1/2	e, μ, τ
neutrino	χ_3	2	0	$+1/2$	$\mathbf{1}$	1/2	ν_1, ν_2, ν_3

All discrete quantum information of fermions—three-generation structure, color charge, electric charge quantization, weak isospin, spin, and Fermi/Bose statistics— has a precise arithmetic counterpart in the BC framework. The three generations arise from the three orbits of the involution; color arises from the resolution of the $\mathbb{Z}/3\mathbb{Z}$ direction in the CRT decomposition; charge quantization arises from the multiplicative decomposition of characters and the equal distribution over color channels; spin arises from the half-order property of the generator. These correspondences involve no continuous parameters or fitting—they are determined entirely by the arithmetic structure of $(\mathbb{Z}/7\mathbb{Z})^*$. The remaining continuous parameters—masses—will be given in §6 by the values of L-functions on the critical line.

4 Jacobi Phase, Cabibbo Angle, and CKM

The preceding two sections established the correspondence between the character group $\widehat{(\mathbb{Z}/7\mathbb{Z})^*}$ and particle species, as well as the description by CRT decomposition of three generations, color, charge, and spin. This section further analyzes the phase relations among characters—how the phase δ_2 of the Jacobi sum $J(\chi_2, \chi_2)$ encodes inter-generation mixing. The Cabibbo angle λ is determined directly by δ_2 (§4.1), and the cyclic structure of inter-generation transitions arises from the three-cycle of the $\times 2$ Hecke action on doublets.

Cyclic structure of inter-generation transitions. In physics, the weak decays among the three generations of quarks (u, c, t) and (d, s, b) form cycles: $u \to c \to t \to u$, $d \to s \to b \to d$. Each transition is accompanied by the emission or absorption of a W boson, and the entire cycle closes among the three generations.

In the BC framework, this cyclic transition is realized by the Hecke-type action of multiplication by 2 on $(\mathbb{Z}/7\mathbb{Z})^*$. Direct calculation of its action on the three doublets:

$$2 \cdot \{1, 6\} = \{2, 12\} \equiv \{2, 5\} = D_2,$$

$$2 \cdot \{2, 5\} = \{4, 10\} \equiv \{4, 3\} = D_3,$$

$$2 \cdot \{3, 4\} = \{6, 8\} \equiv \{6, 1\} = D_1.$$

Therefore

$$D_1 \to D_2 \to D_3 \to D_1,$$

and the Hecke action on the doublet space forms a three-cycle, precisely corresponding to the inter-generation transition cycle in physics.

Theorem 4.1 (Hecke three-cycle)**.** *Let S_m^* denote the Hecke-type action on the BC local factors. For $m = 2$, S_2^* induces a three-cycle on the doublet set $\{D_1, D_2, D_3\}$:*

$$D_1 \to D_2 \to D_3 \to D_1.$$

In particular, its minimal positive order is 3*.*

Proof. The direct calculation above gives $S_2^*(D_1) = D_2$, $S_2^*(D_2) = D_3$, $S_2^*(D_3) = D_1$. Hence $S_2^{*3} = \mathrm{id}$. Since neither S_2^* nor S_2^{*2} is the identity, the minimal positive order is 3. □

The diagonal entries of the CKM matrix correspond to amplitudes remaining within the same doublet, and the off-diagonal entries correspond to amplitudes for transitions between different doublets. Quark mixing is thus embedded in a discrete three-cycle path, rather than an arbitrary 3×3 unitary matrix.

Mixing angle: Jacobi phase δ_2. The *cyclic structure* of inter-generation transitions is determined by the Hecke three-cycle; the *angle* of the transitions requires a separate quantity—the phase of the Jacobi sum.

Physically, the off-diagonal entries of the CKM matrix measure the mixing angles between adjacent generations. In the BC framework, this angle is given by the phase of the Jacobi sum $J(\chi_2, \chi_2) = \tau(\chi_2)^2/\tau(\bar{\chi}_2)$ [15, 25]. The Jacobi sum measures the multiplicative autocorrelation of character χ_2 with itself, and its phase $\arg(J)$ encodes the relative rotation angle among the three-generation doublets. The Eisenstein decomposition gives the exact value:

$$J(\chi_2, \chi_2) = -1 - d_2\omega_3, \quad \omega_3 = e^{2\pi i/3},$$

where $d_2 = 3$. Thus $\arg J = -\arctan(d_2^{3/2}) = -\arctan(3\sqrt{3}) = -79.11°$, giving

$$\delta_2 = \frac{\arg J(\chi_2, \chi_2)}{\varphi(p)} = \frac{-\arctan(d_2^{3/2})}{2d_2} = -13.18°. \tag{6}$$

Modulus–phase separation: the absolute scale $|L(\frac{1}{2}, \chi_2)|^4$ is evaluated at $s = \frac{1}{2}$, the symmetry axis of the functional equation. The CKM mixing angle is determined directly by this bare Jacobi phase $\delta_2 = \arg L(\frac{1}{2}, \chi_2) = -13.18°$. The mass phase used for the charged-lepton spectrum (§6) carries an additional electromagnetic correction: charged leptons bear unit electric charge, so their mass phase is shifted by $-\alpha_{\mathrm{EM}}$ due to the electromagnetic self-energy (Coulomb energy, positive contribution); neutral neutrinos carry no such term,

$$\delta = \arg L(\tfrac{1}{2}, \chi_2) - \alpha_{\mathrm{EM}} = -12.73°,$$

where α_{EM} is evaluated at the electroweak scale. The shift decreases $|\delta|$, consistent with the positive sign of the electromagnetic self-energy; its precise coefficient remains to be derived from the spectral action. The CKM matrix itself uses only δ_2 and does not involve this correction.

4.1 Cabibbo Angle

The experimental Cabibbo angle $\lambda \approx 0.225$ is the largest off-diagonal entry of the CKM matrix, directly measuring the mixing strength between the first and second generation quarks ($d \leftrightarrow s$ transition). In the BC framework, λ is uniquely determined by the Jacobi phase δ_2: the amplitude for a single-step doublet transition is

$$\lambda = \frac{\sin(2\delta_2)}{2}.$$

Substituting $\delta_2 \approx 13.18°$, one obtains

$$\lambda \approx 0.222,$$

in agreement with the experimental value $\lambda_{\mathrm{exp}} \approx 0.2245$ to within one percent.

Proposition 4.2 (Single-step tunneling)**.** *If the minimal inter-doublet transition is induced by the phase difference* $2\delta_2$*, then the single-step tunneling amplitude satisfies*

$$\lambda = \frac{\sin(2\delta_2)}{2} = 0.222,$$

in agreement with the experimental Cabibbo parameter $\lambda_{\mathrm{exp}} \approx 0.2245$ *to the percent level.*

Remark 4.3 (Why $\sin(2\delta_2)/2$?). The functional form $\lambda = \sin(2\delta_2)/2$ has a precise character-theoretic interpretation. Since $\chi_5 = \chi_2 \times \chi_3$ and $\chi_1 = \bar{\chi}_2 \times \chi_3$, the χ_3 factor cancels in the CKM matrix elements; consequently, CKM mixing arises entirely from the interference between χ_2 ($\arg = -\delta_2$) and $\bar{\chi}_2$ ($\arg = +\delta_2$):

- $\sin(\delta_2)$: projection amplitude of the u-type sector onto the mixing direction;
- $\cos(\delta_2)$: projection amplitude of the d-type sector onto the mixing direction;
- product $\sin(\delta_2)\cos(\delta_2) = \frac{1}{2}\sin(2\delta_2)$: superposition of the two projections, i.e., the CKM off-diagonal amplitude.

The factor 1/2 arises from a trigonometric identity, not a free parameter; the specific identification of u/d sectors with sin / cos is a recognition at the amplitude level, and is not uniquely determined by first principles.

5 Arithmetic Structure of the Mass Spectrum

The preceding sections established the correspondence between matter particles and the algebraic structure of the BC system: the character types determine the discrete quantum numbers of fermions (§2–§3), and the Jacobi phase determines the inter-generation mixing angles (§4). This section turns to the arithmetic structure of the mass spectrum—analyzing which factors determine each fermion mass and the arithmetic origin of each factor.

5.1 Mass Operator and L-functions

In the BC framework, fermion masses arise from the non-commutativity of addition and multiplication. Each fermion corresponds to a character direction χ_j, whose mass is measured by the commutator $\mu_j^{(a)} = \langle\chi_j|[D_F, T_a]|\chi_j\rangle$, where T_a is the additive translation operator on $\mathbb{Z}/p\mathbb{Z}$ ($T_a|k\rangle = |k+a\rangle$), and D_F is the finite Dirac operator of the BC spectral triple [12, 16, 17, 29, 30, 34]. When $[D_F, T_a] = 0$, addition and multiplication decouple, and that direction is massless; when $[D_F, T_a] \neq 0$, they do not decouple, and the mass is nonzero. The eigenvalues of D_F in the character basis are proportional to the Gauss sum $\tau(\chi_j)$:

$$\mu_j^{(a)} = \langle\chi_j|[D_F, T_a]|\chi_j\rangle. \tag{7}$$

The entanglement of the real character (χ_3) cancels exactly at tree level, giving a zero diagonal entry—corresponding to neutrinos; the entanglement of complex characters (χ_2, χ_1) does not self-cancel—corresponding to massive leptons and quarks.

The above commutator $\mu_j^{(a)}$ is a local quantity defined on $\mathbb{Z}/p\mathbb{Z}$. Locally, the entanglement strength of all nontrivial characters is identical: the Gauss sum satisfies $|\tau(\chi_j)| = \sqrt{p}$— if one only looks at $\mathbb{Z}/7\mathbb{Z}$, all fermion masses are equal with no hierarchy. The mass hierarchy arises from propagation from local to global: the entanglement seed on $\mathbb{Z}/p\mathbb{Z}$ is propagated to all of $\mathbb{Z}$, and different characters are modulated differently by all primes $q = 2, 3, 5, \ldots$, yielding the value of the Dirichlet L-function [22] on the critical line $s = 1/2$:

$$L\left(\tfrac{1}{2}, \chi_j\right) = \sum_{n=1}^{\infty} \frac{\chi_j(n)}{\sqrt{n}}.$$

$s = 1/2$ is the symmetry axis of the functional equation $\Lambda(s, \chi) = \Lambda(1-s, \bar{\chi})$, and corresponds to the KMS phase transition point $\beta = 1$ of the BC system ($s = \beta/2$). $|L(\frac{1}{2}, \chi_j)|$ is no longer equal for different j: $|L_2| = 0.31848$, $L_3 = 1.14659$, $|L_1| = 0.85747$ (L_3 has no absolute value because χ_3 is a real character, $L(\frac{1}{2}, \chi_3) \in \mathbb{R}$).

5.2 Subfield Correspondence

The masses of different fermions involve different combinations of L-functions. This combination is not assigned arbitrarily, but is uniquely determined by the cyclic subgroup generated by the character of that fermion.

Let the character of fermion f be χ_f, and denote by $\langle\chi_f\rangle \subseteq \widehat{(\mathbb{Z}/7\mathbb{Z})^*}$ the cyclic subgroup it generates. By class field theory, $\langle\chi_f\rangle$ corresponds to a unique intermediate subfield K_f satisfying $\mathbb{Q} \subseteq K_f \subseteq \mathbb{Q}(\zeta_7)$, and the Artin factorization gives

$$\prod_{\chi\in\langle\chi_f\rangle} L(s, \chi) \;=\; \frac{\zeta_{K_f}(s)}{\zeta(s)}. \tag{8}$$

Therefore, the mass scale of fermion f is proportional to the value of its subfield's Dedekind zeta on the critical line:

$$m_f^{\text{scale}} \;\propto\; \left(\frac{\zeta_{K_f}(\frac{1}{2})}{\zeta(\frac{1}{2})}\right)^2,$$

where the square comes from the Born product of the emission and absorption vertices. Checking sector by sector:

- **Leptons**: coupled to χ_2 (order 3, $\mathbb{Z}/3\mathbb{Z}$ direction). The generated subgroup $\langle\chi_2\rangle = \{\chi_0, \chi_2, \chi_4\}$ corresponds to the totally real subfield $\mathbb{Q}(\zeta_7)^+$. L-functions involved: only L_2.
- **Up-type quarks**: character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ has order 6, generating the full character group. Corresponds to the full cyclotomic field $\mathbb{Q}(\zeta_7)$. Involves all three L-functions: L_1, L_2, L_3.
- ***d*-type quarks**: corresponds to the set of quadratic non-residues $\mathrm{QNR} = \{3, 5, 6\}$. This set is not a subgroup (not closed under multiplication: $\mathrm{QNR} \cdot \mathrm{QNR} = \mathrm{QR}$), so the d-type spans two subfields: starting from the lepton subfield $\mathbb{Q}(\zeta_7)^+$, passing through $\mathbb{Q}(\sqrt{-7})$. Involves: L_2, L_3.
- **Neutrinos**: corresponds to χ_3 (order 2, the unique real character). The generated subgroup $\langle\chi_3\rangle = \{\chi_0, \chi_3\}$ corresponds to the quadratic subfield $\mathbb{Q}(\sqrt{-7})$. Involves: L_3.

Summary table:

sector	$\langle\chi_f\rangle$	Subfield K_f	L-functions involved
Leptons	$\langle\chi_2\rangle$ (order 3)	$\mathbb{Q}(\zeta_7)^+$ (totally real cubic)	L_2
d-type quarks	QNR set	spans $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\sqrt{-7})$	L_2, L_3
u-type quarks	$\langle\chi_1\rangle$ (order 6)	$\mathbb{Q}(\zeta_7)$ (full)	L_1, L_2, L_3
Neutrinos	$\langle\chi_3\rangle$ (order 2)	$\mathbb{Q}(\sqrt{-7})$ (quadratic)	L_3

5.3 Propagation Paths

The subfield correspondence determines which L-functions are involved for each fermion. The next step is to translate the subfield correspondence into concrete **propagation paths**— the sequence of nodes traversed as the seed propagates from the vacuum (χ_0) through character space to reach the target direction. This is precisely the notion of "path" used extensively in §6.

The rule from subfield to path: *The character nodes visited by the path = the nontrivial directions in the character subgroup of that fermion (modulo Galois conjugation).* Propagation starts from the vacuum (χ_0) in character space and passes successively through the character directions contained in the target subfield. For complex characters, the direct channel is open, and the subfield correspondence gives the path directly; for the real character (χ_3), the direct channel is closed (tree-level mass is zero), and a nonzero mass is obtained via the indirect path ($\chi_3^2 = \chi_0$, routing through the vacuum).

Applying sector by sector:

- **Leptons**: $\langle\chi_2\rangle$ has only one independent direction χ_2 (conjugate χ_4 is not independent). Path: $\chi_0 \to \chi_2$. One step, shortest.
- ***d*-type quarks**: the QNR set spans two directions χ_2 and χ_3. Path: $\chi_0 \to \chi_2 \to \chi_3$. Two steps.
- **Up-type quarks**: $\langle\chi_1\rangle$ is the full group, containing three independent directions χ_2, χ_3, χ_1. Path: $\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$. Three steps, longest.
- **Neutrinos**: $\langle\chi_3\rangle$ has only the χ_3 direction, but χ_3 is a real character, so the direct channel is closed. $\chi_3^2 = \chi_0$ gives the unique indirect channel: the signal propagates through χ_0 (vacuum/KMS heat bath) and back, passing through χ_2 twice. Path: $\chi_0 \to \chi_2 \to [\chi_0 \text{ round trip}] \to \chi_2$.

Path summary:

sector	path	L-functions
Leptons	$\chi_0 \to \chi_2$	L_2
d-type quarks	$\chi_0 \to \chi_2 \to \chi_3$	L_2, L_3
u-type quarks	$\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$	L_1, L_2, L_3
Neutrinos	$\chi_0 \to \chi_2 \to [\chi_0 \text{ round trip}] \to \chi_2$	L_2

The path length determines the complexity of the mass formula: the more character nodes traversed, the more types of L-factors, and the more intricate the node normalization and generation corrections.

Power rule. The subfield and path determine which L-functions are involved for each fermion, but the same L-function may appear with different powers in the formulas for different fermions. The difference in powers arises from whether the two pairing structures of the character degenerate— when degenerate, propagation is at the amplitude level (lower power); when non-degenerate, at the probability level (higher power).

Specifically, the power of each L-factor = Galois degree × coherence:

Galois × Coherence	Coherent (C=1)	Independent (C=2)
Complex character (G=2)	2	4
Real character (G=1)	1	2

Two criteria:

- **Galois degree** G: complex characters have conjugate pairs $\chi_k \leftrightarrow \bar{\chi}_k$ (two states), $G = 2$; real characters are self-conjugate (degenerate to one state), $G = 1$.
- **Coherence** C: group elements have multiplicative inverse pairs $a \leftrightarrow a^{-1}$. Self-inverse ($a^2 = 1$) degenerates to one state, coherent propagation, $C = 1$; non-self-inverse gives two independent states, Born probability level, $C = 2$. This criterion applies to the *terminal* character of the fermion (the sector where the fermion resides). Characters traversed in the interior of the path all take $C = 2$: when the signal passes through a non-resident sector it propagates at the Born probability level, not triggering coherent degeneracy.

The power of L_3 for down-type quarks varies by generation: the d-type corresponds to the QNR set $\{3, 5, 6\}$, and the self-inverse property of each element differs:

d-type quark	QNR element a	a^2 mod 7	Self-inverse?	L_3 power
b (gen3)	$6 = -1$	1	Yes	L_3^1
s (gen2)	5	4	No	L_3^2
d (gen1)	3	2	No	L_3^2

Leptons: $\chi_0 \to \chi_2$. Leptons couple only to the even character χ_2 (order 3, $\mathbb{Z}/3\mathbb{Z}$ direction). One-step propagation; the mass formula contains only L_2:

$$m_\ell \propto |L_2|^4.$$

The power $4 = G \times C = 2 \times 2$: complex character ($G = 2$), the unique nontrivial direction in the lepton sector ($C = 2$, independent Born). This is the shortest path and the simplest mass formula.

d-type quarks: $\chi_0 \to \chi_2 \to \chi_3$. d-type quarks (b, s, d) additionally pass through χ_3 (order 2, $\mathbb{Z}/2\mathbb{Z}$ direction, weak-force sector) beyond the lepton path. Two-step propagation; the mass formula contains both L_2 and L_3:

$$m_d \propto |L_2|^4 \cdot L_3^{\text{power}}.$$

The power of L_3 varies by generation: gen3 (b) is power 1 (coherent, $C = 1$); gen2/gen1 (s, d) is power 2 (independent Born, $C = 2$). The mass hierarchy of down-type quarks therefore has an additional modulation by an L_3 factor compared to leptons.

Up-type quarks: $\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$. The character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ of up-type quarks (c, t, u) encodes the information of all nontrivial characters. Three-step propagation; the mass formula contains all three L-values:

$$m_{u\text{-type}} \propto |L_2|^4 \cdot L_3^2 \cdot |L_1|^4.$$

The longest path and most complex formula. $|L_1|^4$ likewise comes from the Born probability for complex characters ($G \times C = 2 \times 2$). Up-type quarks "see" the full $\mathbb{Z}/6\mathbb{Z}$ structure; this is the arithmetic reason why their inter-generation structure involves force coupling constants (rather than purely phase projections). The L-factors are identical for all three generations; all mass differences among the three generations come from generation factors, with the Dyson self-loop of the top quark and the weak coupling of the u quark treated in the corresponding subsections.

Neutrinos: $\chi_0 \to \chi_2 \to [\chi_0$ round trip$] \to \chi_2$. The character χ_3 of neutrinos is real ($\chi_3 = \bar{\chi}_3$, order 2). The diagonal commutator of a real character is zero at tree level: the direct channel is closed. A nonzero mass must be achieved via an indirect path.

The character multiplication $\chi_3^2 = \chi_0$ gives the unique indirect channel: the signal from χ_3 is mapped to the trivial character χ_0 via squaring, and then returns from χ_0 to χ_3. Physically, this corresponds to the neutrino field completing one Majorana-type round trip via the χ_0 channel (vacuum/KMS heat bath)— the signal must "route through" the vacuum in order to propagate, because the neutrino's own channel has no propagating mode. χ_0 is the trivial character, carrying $N = \varphi(p^K) = \varphi(7^4) = 2058$ vacuum modes. $K = 4$ is determined by the color structure: the CRT decomposition gives $n_c = 3$ independent color directions, the p-adic expansion of $(\mathbb{Z}/p^K\mathbb{Z})^*$ provides $K - 1$ independent $\mathbb{F}_p$ coordinates, one for each color direction, so $K - 1 = n_c = 3$, $K = 4$. The signal is diluted among N modes upon entry, and again diluted upon return.

The L-factor doubles due to the round trip:

$$m_\nu \propto |L_2|^8.$$

The lepton path gives $|L_2|^4$; the χ_0 round trip multiplies by another $|L_2|^4$, yielding $|L_2|^8$ in total.

Path length determines the complexity of the mass formula: the more character nodes traversed, the more types of L-factors and node normalization factors. Leptons pass through only χ_2 (one step); up-type quarks pass through all three nontrivial characters (three steps). The L-factors of neutrinos share the same origin as those of leptons (both are powers of $|L_2|$), but the N^2 dilution factor introduced by the χ_0 round trip ($N^2 \approx 4.2\times10^6$) together with the additional power of $|L_2|^4$ jointly suppress the neutrino mass to the meV scale.

5.4 Node Factors

The path determines the L-factors in the mass formula. The mass of each fermion also requires the normalization weights of each character node along the path—the node factors. The node factors are determined by the algebraic structure of the CRT decomposition $\mathbb{Z}/6\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, physically corresponding to the principle: *indistinguishable (symmetric) states are normalized at amplitude level by a factor $1/\sqrt{n}$; distinguishable (independent) states are summed at probability level with multiplicity n.*

Upon reaching character χ_k, three independent effects may be triggered:

1. **Galois projection**: complex characters (χ_2, χ_1) have Galois pairs $(\chi_k \leftrightarrow \bar{\chi}_k)$, physically indistinguishable—symmetric states, amplitude-level $\div\sqrt{d_1}$, $d_1 = [\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7)^+] = 2$. Real character (χ_3) is self-conjugate, no projection, $\times 1$.
2. **Color activation**: χ_3 corresponds to the $\mathbb{Z}/2\mathbb{Z}$ direction in the CRT decomposition; the three color states are independent in this direction—distinguishable states, probability-level $\times d_2 = \times 3$. Other characters do not trigger this.
3. **Background normalization**: when the path passes from χ_2 into χ_3, the contribution $\zeta(\frac{1}{2})$ of the trivial character χ_0 provides the baseline; the crossing cost is $\div|\zeta(\frac{1}{2})|$. Passing through χ_0 itself results in dilution over $N = \varphi(p^4) = 2058$ KMS-active modes, at cost $\div N$ per traversal.

χ_k	Galois projection	Color activation	Background norm.	Node factor
χ_2	$\div\sqrt{d_1}$	$\times 1$	—	$1/\sqrt{2}$
χ_3	$\times 1$	$\times d_2$	$\div\lvert\zeta\rvert$	$3/1.46035$
χ_1	$\times 1$	$\times 1$	—	1
χ_0	—	—	$\div N$	$1/2058$

Node factor application table. The node factor for each fermion = the single-traversal factor at each node along the path $\times$ number of traversals:

Particle	χ_2 $(\div\sqrt{d_1})$	χ_3 $(\times d_2/\lvert\zeta\rvert)$	χ_1 $(\times 1)$	χ_0 $(\div N)$	Node factor
τ, μ, e	$\times 1$	—	—	—	$1/\sqrt{d_1}$
b	$\times 1$	$\times 1$	—	—	$d_2/(\sqrt{d_1}\,\lvert\zeta\rvert)$
s, d	$\times 1$	$\times 1$ (Born)	—	—	$1/\lvert\zeta\rvert^2$
c, t, u	$\times 1$	$\times 1$	$\times 1$	—	$1/\sqrt{d_1}$
ν	$\times 2$	—	—	$\times 2$	$d_1^{-1}N^{-2}$

ν passes through χ_2 twice $((\div\sqrt{d_1})^2 = \div d_1)$ and χ_0 twice $((\div N)^2 = \div N^2)$. For s/d, the χ_3 traversal is an independent Born mode (C=2), and the color activation d_2 is consumed in the second traversal.

The b quark (gen3, doublet D_1 with the largest tension) shares K_0 with lepton τ, and maintains coherence (amplitude-level propagation, C=1) when crossing the sector at χ_3—corresponding to a single color activation $\times d_2$ and a single background normalization $\div|\zeta|$. For s, d quarks (gen2/gen1), the generation factor requires an additional phase/Cabibbo rotation, and the cross-sector traversal degenerates to an independent Born process (C=2)—the color activation factor is absorbed into the Born square, and the node simplifies to $1/|\zeta|^2$.

5.5 Generation Factors

The path determines the L-factors in the mass formula, and the node factors assign normalization weights. The generation factor is the third component of the mass formula, measuring the contribution of the $\mathbb{Z}/3\mathbb{Z}$ phase structure to the three-generation masses, and is determined by the algebraic structure of the $\mathbb{Z}/3\mathbb{Z}$ component of the CRT decomposition.

For the three generations of particles arriving at the terminal point of the path, two independent effects contribute to the mass splitting:

First layer: phase projection. The three doublets D_1, D_2, D_3 correspond to the three elements of $\mathbb{Z}/3\mathbb{Z}$, evenly spaced on the phase circle (separated by 120°). The Jacobi phase $\delta_2 = -13.18°$ does not coincide with any lattice point, so its cos projections onto the three lattice points are unequal:

$$f_k = 1 + \sqrt{d_1}\cos(\delta + 120° k), \qquad k = 0, 1, 2.$$

The basic generation factor is

$$K_k = (f_k/f_0)^2.$$

The parameter $\sqrt{d_1} = \sqrt{2}$ comes from the Galois degree (same as §5.4). $K_0 \equiv 1$ labels the **reference generation**—the generation with the largest projection and no additional generation correction. Phase projection applies to all charged fermions: the three generations of each sector are labeled by K_0, K_1, K_2. Rule for identifying the reference generation: in the absence of second-layer virtual process corrections, the reference generation is the heaviest generation. For leptons and d-type quarks, the reference generation is gen3 (τ, b); for up-type quarks, gen3 (t) is shifted from the phase projection baseline by the Dyson enhancement, and gen2 (c) becomes the reference generation.

For the lepton sector with only a one-step path, there is no intermediate node to trigger virtual processes, so the generation factor equals pure phase projection. For d-type and up-type quarks with longer paths, the second-layer virtual process corrections will append additional factors on top of the phase projection.

Second layer: virtual process correction (transition series). When the path has two or more steps, intermediate character nodes can act as transition nodes, producing additional corrections to the inter-generation propagation. The correction factors are uniquely determined by the **topology of the transition channel**. The key criterion is whether the character produced after the transition returns to the original direction:

- **Closed channel**: the transition product returns to the original direction (or its Galois conjugate), allowing infinite iteration. Geometric series summation gives a factor $1/\alpha$.
- **Open channel**: the transition product leaves the original direction, and cannot be iterated. The single-transition probability gives a factor α.

Checking direction by direction:

1. χ_2 **direction (electromagnetic)**: $\chi_2^2 = \chi_4 = \bar{\chi}_2$. The transition product is the Galois conjugate of χ_2, physically indistinguishable from χ_2, forming a closed channel. Summing the geometric series $\sum_{n=0}^{\infty} \alpha_{\mathrm{EM}}^n$ gives a factor $1/\alpha_{\mathrm{EM}} \approx 137$.
2. χ_3 **direction (weak force)**: $\chi_3^2 = \chi_0 \neq \chi_3$. The transition product is the trivial character χ_0, which does not return to the χ_3 direction, forming an open chan-

nel. The single-transition probability gives a factor $\alpha_W = g_2^2/(4\pi) \approx 1/29.6$ (weak coupling constant evaluated at M_Z).

3. $\mathbb{Z}/3\mathbb{Z}$ **direction (inter-generation)**: A transition from lattice point k to the adjacent point $k+1$ does not return to k, forming an open channel. The single-step transition probability $\lambda^2 = [\sin(2\delta_2)/2]^2 \approx 0.049$ shares the same Jacobi phase δ_2 as the Cabibbo angle in §4.

Triggering conditions. Virtual process corrections appear only when there exist activatable transition nodes along the path. The lepton path has only one step with no intermediate nodes, so the generation factor is pure phase projection. The d-type path has two steps ($\chi_2 \to \chi_3$), where gen1 requires an additional jump of one $\mathbb{Z}/3\mathbb{Z}$ lattice point from gen2, activating the inter-generation direction. The u-type path has three steps ($\chi_2 \to \chi_3 \to \chi_1$), and the character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ encodes information from both the electromagnetic and weak-force directions, allowing the transition channels of both directions to be activated.

Generation factor composition table.

Direction	**Channel topology**	**Transition type**	**Factor**
χ_2 (EM)	closed ($\chi_2^2 = \bar{\chi}_2$)	Dyson series	$1/\alpha_{\mathrm{EM}}$
χ_3 (weak)	open ($\chi_3^2 = \chi_0$)	single transition	α_W
$\mathbb{Z}/3\mathbb{Z}$ (inter-gen)	lattice hop	single-step probability	λ^2

Generation factor application table.

Particle	K_k	$\mathbb{Z}/3\mathbb{Z}$	χ_2	χ_3	Total Gen
τ	K_0	—	—	—	K_0
μ	K_1	—	—	—	K_1
e	K_2	—	—	—	K_2
b	K_0	—	—	—	K_0
s	K_1	—	—	—	K_1
d	K_1	$\times\lambda^2$	—	—	$K_1\lambda^2$
c	K_0	—	—	—	K_0
t	K_0	—	$\times 1/\alpha_{\mathrm{EM}}$	—	K_0/α_{EM}
u	K_0	$\times\lambda^2$	—	$\times\alpha_W$	$K_0\lambda^2\alpha_W$

The real character χ_3 of neutrinos is self-conjugate ($\chi_3 = \bar{\chi}_3$), locking the phase, so phase projection is not supported. Their generation factor is replaced by the eigenvalues λ_k of the cyclic-Hurwitz mass matrix (§6.5).

6 Mass Formulas

The preceding section established the arithmetic structure of the mass spectrum. There are three independent sources of fermion mass differences: different fermions correspond to different subfields, traverse different paths, and involve different combinations of L-functions—this determines the inter-sector mass scale differences; the CRT symmetry properties of each character node on the path determine its normalization weight; the splitting of three generations within the same sector arises from the unequal projections of the Jacobi phase δ_2 at the three lattice sites of $\mathbb{Z}/3\mathbb{Z}$. The three factors are independent and combine multiplicatively to yield each fermion's mass. The unified formula is presented below and expanded for each fermion class.

6.1 Unified Mass Formula

Combining the path, node, and generation factors, each fermion mass is assembled from three factors:

$$\boxed{m_f = v \times \underbrace{\prod_{\text{path}} |L_k|^{\text{power}}}_{\text{scale}} \times \underbrace{\prod_{\text{nodes}} \text{normalization}}_{\text{node factor}} \times \underbrace{\text{Gen}_f}_{\text{generation factor}} .} \tag{9}$$

$v = 246.22\,\text{GeV}$ is the experimentally measured Higgs VEV.

Physical interpretation of the three factors.

- **Scale** (L-factor): the Born probability after the local conflict seed propagates globally on $\mathbb{Z}$. The path determines which characters are traversed; the power rules determine the Born level at each step. The inter-sector mass scale differences arise entirely from the unequal moduli of $|L_2|, L_3, |L_1|$ at $s = 1/2$.
- **Node factor**: the normalization weight assigned to each node by the CRT structure of the character space. The Galois symmetry ($\div\sqrt{d_1}$), color degrees of freedom ($\times d_2$), background baseline ($\div|\zeta|$), and KMS dilution ($\div N$) are all uniquely determined by the algebraic structure, with no free parameters.
- **Generation factor** (Gen): the contribution of the $\mathbb{Z}/3\mathbb{Z}$ phase structure to the three-generation masses. Phase projection gives the basic splitting; virtual process transitions along each direction of the path— closed loops yield $1/\alpha$, open channels yield α—provide additional corrections. See §5.5 for details.

The three factors are uniquely determined by the path (§5.3), the CRT structure (§5.4), and the $\mathbb{Z}/3\mathbb{Z}$ phase (§5.5), respectively, with no free parameters. Below we expand the unified formula for each of the four fermion classes and give the specific mass of each generation.

6.2 Lepton Masses

The lepton path is $\chi_0 \to \chi_2$ (§5.3); the unified formula expands to:

$$m_\ell = v \times \underbrace{|L_2|^4}_{L\text{-factor}} \times \underbrace{\frac{1}{\sqrt{d_1}}}_{\text{node}} \times \underbrace{K_k}_{\text{gen}} .$$

L-factor: $|L_2|^4$. Leptons couple only to the even character χ_2; the mass scale is determined by the modulus of $L(\frac{1}{2}, \chi_2)$. $|L_2|^4$ is the Born probability of the complex character ($G \times C = 2 \times 2$, the unique solution from C^*-algebra positivity and normalization [23]). The Dedekind decomposition gives its location in the cyclotomic field $K = \mathbb{Q}(\zeta_7)$:

$$y_\tau = |L(\tfrac{1}{2}, \chi_2)|^4 = \left(\frac{\zeta_{K^+}(\frac{1}{2})}{\zeta(\frac{1}{2})}\right)^2,$$

i.e., the share of the totally real subfield K^+ on the critical line.

Node factor: $1/\sqrt{d_1}$**.** The Galois pair of χ_2 ($\chi_2 \leftrightarrow \bar{\chi}_2$) is physically indistinguishable; the symmetric state is normalized at the amplitude level by $\div\sqrt{d_1}$, where $d_1 = [\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7)^+] = 2$.

Generation factor: K_k**.** The three-generation mass ratios are given by the *phase* δ of $L(\frac{1}{2}, \chi_2)$ via phase projection onto $\mathbb{Z}/3\mathbb{Z}$:

$$f_k = 1 + \sqrt{d_1}\cos(\delta + 120°k), \qquad K_k = (f_k/f_0)^2.$$

$\delta = -12.73°$ (the Jacobi phase δ_2 shifted by the electromagnetic correction $-\alpha_{\mathrm{EM}}$, see §4). $r = \sqrt{d_1} = \sqrt{2}$ comes from the Galois degree (§5.4).

Three-generation mass formulas.

$$m_\tau = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_0 = \frac{|L_2|^4}{\sqrt{2}} \times v, \tag{10}$$

$$m_\mu = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_1 = m_\tau \times (f_1/f_0)^2, \tag{11}$$

$$m_e = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_2 = m_\tau \times (f_2/f_0)^2. \tag{12}$$

$|L_2| = 0.31848$ (scale) and $\delta = -12.73°$ (phase) control the absolute mass and inter-generation ratios, respectively.

Particle	L-factor	Node	Gen	Prediction	Experiment	Deviation
τ	$\|L_2\|^4$	$1/\sqrt{d_1}$	K_0	1791 MeV	1777 MeV	+0.8%
μ	$\|L_2\|^4$	$1/\sqrt{d_1}$	K_1	106.5 MeV	105.7 MeV	+0.8%
e	$\|L_2\|^4$	$1/\sqrt{d_1}$	K_2	0.5140 MeV	0.5110 MeV	+0.7%

All three-generation deviations are $< 1\%$. The extreme hierarchy $m_\tau/m_e \approx 3500$ arises from the near-zero crossing $f_2 \to 0$.

6.3 d-type Quark Masses

The d-type quark path is $\chi_0 \to \chi_2 \to \chi_3$ (§5.3), corresponding to the Georgi–Jarlskog relation [4]. The unified formula expands to:

$$m_d = v \times \underbrace{|L_2|^4 \cdot L_3^{\mathrm{pow}}}_{L\text{-factor}} \times \underbrace{\text{node}}_{\text{see below}} \times \underbrace{\text{Gen}}_{\text{gen}}.$$

L**-factor:** $|L_2|^4 \cdot L_3^{\mathbf{pow}}$**.** Building on the leptonic $|L_2|^4$, d-type quarks additionally traverse χ_3 (the weak-force sector). $L_3 = L(\frac{1}{2}, \chi_3) = 1.14659$ is a real character, self-conjugate ($\chi_3 = \bar{\chi}_3$). The power of L_3 depends on the generation: the b quark (gen3) shares the χ_2 direction with leptons, and the cross-sector traversal is amplitude-level $\to L_3^1$; for s, d quarks the χ_3 traversal is an independent Born process $\to L_3^2$.

Node factor. b quark: the path traverses χ_2 (Galois $\div\sqrt{d_1}$) and χ_3 (color activation $\times d_2$, background normalization $\div|\zeta|$), giving node $= d_2/(\sqrt{d_1}\,|\zeta|) = 3/(\sqrt{2} \times 1.46035)$. s, d quarks: the color activation d_2 at χ_3 is consumed during the second traversal (Born level), giving node $= 1/|\zeta|^2$.

Generation factor. b (gen3) uses K_0; s (gen2) uses K_1; d (gen1) uses K_1 multiplied by Cabibbo λ^2. The generation factor of the d quark is $K_1\lambda^2$: first projected to gen2 by phase projection (K_1), then suppressed by the Cabibbo hop (λ^2).

Three-generation mass formulas.

$$m_b = v \times |L_2|^4 L_3 \times \frac{d_2}{\sqrt{d_1}\,|\zeta|} \times K_0, \tag{13}$$

$$m_s = v \times |L_2|^4 L_3^2 \times \frac{1}{|\zeta|^2} \times K_1, \tag{14}$$

$$m_d = v \times |L_2|^4 L_3^2 \times \frac{1}{|\zeta|^2} \times K_1\lambda^2. \tag{15}$$

Particle	L-factor	Node	Gen	Prediction	Experiment	Deviation
b	$\|L_2\|^4 L_3$	$d_2/(\sqrt{d_1}\,\|\zeta\|)$	K_0	4219 MeV	4180 MeV	+0.9%
s	$\|L_2\|^4 L_3^2$	$1/\|\zeta\|^2$	K_1	92.90 MeV	93.40 MeV	−0.5%
d	$\|L_2\|^4 L_3^2$	$1/\|\zeta\|^2$	$K_1\lambda^2$	4.580 MeV	4.670 MeV	−1.9%

All three-generation deviations are $\leq 2\%$. The b/lepton ratio $m_b/m_\tau = d_2 L_3/|\zeta| = 2.356$ (experiment 2.352, deviation +0.1%) equals the color number times the L-function ratio.

6.4 Up-type Quark Masses

The up-type quark propagation path is $\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$ (§5.3); the unified formula expands to:

$$m_u = v \times \underbrace{|L_2|^4 \cdot L_3^2 \cdot |L_1|^4}_{L\text{-factor}} \times \underbrace{\frac{1}{\sqrt{d_1}}}_{\text{node}} \times \underbrace{\text{Gen}}_{\text{gen}}.$$

L-factor: $|L_2|^4 L_3^2 |L_1|^4$. The longest path, containing all three L-values. Both $|L_2|^4$ and $|L_1|^4$ are Born probabilities for complex characters ($G \times C = 2 \times 2$); L_3^2 is the independent Born probability for the real character ($G \times C = 1 \times 2$). The character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ of up-type quarks encompasses all nontrivial characters, so up-type quarks "see" the full $\mathbb{Z}/6\mathbb{Z}$ structure. Unlike the amplitude-level L_3^1 of d-type quarks, the up-type and lepton characters differ ($\chi_1 \neq \chi_2$), so the cross-sector factor degenerates to the Born probability level.

Node factor: $1/\sqrt{d_1}$. The path traverses χ_2 (Galois $\div\sqrt{d_1}$), χ_3, and χ_1. The χ_3 and χ_1 nodes are transparent (§5.4), so the total node factor $= 1/\sqrt{d_1} = 1/\sqrt{2}$, same as for leptons.

Generation factor. The character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ of up-type quarks simultaneously encodes both the electromagnetic and weak-force directions, and the virtual process channels (§5.5) of both directions can be activated:

- c (gen2, reference generation): K_0, no additional coupling;
- t (gen3): closed channel in the χ_2 direction ($\chi_2^2 = \bar{\chi}_2$), Dyson enhancement $1/\alpha_{\text{EM}}$;
- u (gen1): open channel in the χ_3 direction ($\chi_3^2 = \chi_0$) incurs cost α_W, combined with inter-generation hop λ^2.

Three-generation mass formulas.

$$m_c = v \times |L_2|^4 L_3^2 |L_1|^4 \times \frac{1}{\sqrt{d_1}} \times K_0, \tag{16}$$

$$m_t = v \times |L_2|^4 L_3^2 |L_1|^4 \times \frac{1}{\sqrt{d_1}} \times \frac{K_0}{\alpha_{\mathrm{EM}}}, \tag{17}$$

$$m_u = v \times |L_2|^4 L_3^2 |L_1|^4 \times \frac{1}{\sqrt{d_1}} \times K_0 \lambda^2 \alpha_W. \tag{18}$$

The L-factors and node factors are identical for all three generations; all differences lie in the generation factor.

Particle	L-factor	Node	Gen	Prediction	Experiment	Deviation
c	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	K_0	1273 MeV	1270 MeV	+0.2%
t	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	K_0/α_{EM}	174.5 GeV	172.8 GeV	+1.0%
u	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	$K_0\lambda^2\alpha_W$	2.119 MeV	2.160 MeV	−1.9%

All three-generation deviations are $\leq 2\%$. $m_t/m_c = 1/\alpha_{\mathrm{EM}} \approx 137$: the anomalously large top mass is a direct consequence of the EM self-loop. $m_u/m_c = \lambda^2\alpha_W \approx 1/600$: the extreme lightness of the u quark is the result of Cabibbo suppression $\times$ weak-force suppression.

6.5 Neutrino Masses

The neutrino propagation path is $\chi_0 \to \chi_2 \to [\chi_0$ round trip$] \to \chi_2$ (§5.3); the unified formula expands to:

$$m_\nu = v \times \underbrace{|L_2|^8}_{L\text{-factor}} \times \underbrace{\frac{1}{d_1 N^2}}_{\text{node}} \times \underbrace{\lambda_k}_{\text{gen}}.$$

L-factor: $|L_2|^8$. The χ_0 round trip doubles the $|L_2|^4$ of the lepton path. Physically, the character χ_3 of neutrinos is real ($\chi_3 = \bar{\chi}_3$), so the direct channel is closed. $\chi_3^2 = \chi_0$ provides the unique indirect channel: the signal propagates through χ_0 and back, equivalent to traversing the lepton path once more, giving $|L_2|^4 \times |L_2|^4 = |L_2|^8$.

Node factor: $1/(d_1 N^2)$. χ_2 is traversed twice: Galois projection $(\div\sqrt{d_1})^2 = \div d_1$. χ_0 is traversed twice: KMS dilution $(\div N)^2 = \div N^2$, $N = \varphi(p^4) = 2058$. Total node factor $= 1/(d_1 N^2)$. This is the primary reason for the extreme smallness of neutrino masses: dilution across $N^2 \approx 4.2 \times 10^6$ channels.

Generation factor: Hurwitz weights λ_k. The neutrino generation factor is not given by K_k, but is determined by the eigenvalues of the cyclic-Hurwitz mass matrix (unlike the traditional seesaw mechanism [10, 11, 20, 27], no high-energy scale M_R is introduced here).

The doublet product $V_k = a_k(p{-}a_k)$ satisfies the Fermat condition $V_k^3 \equiv -1 \pmod 7$—where -1 is the involution, the arithmetic expression of the Majorana condition, and $n = 3 = (p-1)/2$ is the minimal unifying exponent.

The Hurwitz ζ-function lifts the local Fermat structure to the global level:

$$w_k = \frac{1}{p^6}\,\zeta_H\!\left(3,\,\frac{a_k}{p}\right) \times \zeta_H\!\left(3,\,\frac{p-a_k}{p}\right).$$

The circulant matrix $C = \mathrm{circ}(f_0, f_2, f_1)$ combined with the Hurwitz weights gives the mass matrix:

$$\mathcal{M} = C \cdot \mathrm{diag}(w_1, w_2, w_3) \cdot C^T.$$

C shares the same parameters as the charged leptons ($r = \sqrt{2}$, phase δ), but $w_1 \neq w_2 \neq w_3$ breaks the $\mathbb{Z}/3\mathbb{Z}$ cyclic symmetry, causing the eigenvalues λ_k to deviate from the simple K_k.

Three-generation mass formulas.

$$m_k = v \times |L_2|^8 \times \frac{1}{d_1 N^2} \times \lambda_k, \qquad k = 1, 2, 3. \tag{19}$$

Numerical values: $\lambda_1 = 1.660$, $\lambda_2 = 3.250$, $\lambda_3 = 16.45$; scale $= 3.080\,\mathrm{meV}$. Here $\lambda_k = (\varepsilon_k/2) \times 10^3$, where ε_k are the eigenvalues of $\mathcal{M}$; the factor 1/2 is the Majorana symmetry factor ($\chi_3 = \bar{\chi}_3$ renders the round-trip path indistinguishable).

Particle	L-factor	Node	Gen	Prediction	Experiment	Deviation
ν_3	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	λ_3	50.67 meV	∼50 meV	+1.3%
ν_2	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	λ_2	10.01 meV	—	—
ν_1	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	λ_1	5.113 meV	—	—

$\Sigma m_\nu = 65.79\,\mathrm{meV}$, within CMB-S4 sensitivity ($\sim 30\,\mathrm{meV}$) [24]. $R = \Delta m^2_{31}/\Delta m^2_{21} = 34.31$, consistent with experiment 34.0 ± 1.4 [9] (0.2σ).

CP phase. The CP-violating phase of neutrinos is likewise determined by the arithmetic properties of χ_3. The Gauss sum $\tau(\chi_3) = i\sqrt{7}$ is purely imaginary; its argument locks $\delta_{CP} = -\arg(\tau(\chi_3)) = -\pi/2 = -90°$, consistent with the NOvA/T2K best-fit values [7, 8].

7 Discussion and Predictions

§2–5 have demonstrated that the 12 fermion masses are completely determined by the character structure of the BC system, the Jacobi phase, and the values of the Dirichlet L-functions on the critical line, with all deviations $\leq 2\%$.

7.1 Summary Table

The following collects all fermion mass predictions. Each mass is given by the unified formula (§6.1): $m_f = v \times |L|^{\mathrm{pow}} \times \mathrm{node} \times \mathrm{Gen}$.

Particle	Path	L-factor	Node	Gen	Prediction	Experiment	Deviation
τ	$\chi_0 \to \chi_2$	$\lvert L_2\rvert^4$	$1/\sqrt{d_1}$	K_0	1791 MeV	1777	+0.8%
μ	$\chi_0 \to \chi_2$	$\lvert L_2\rvert^4$	$1/\sqrt{d_1}$	K_1	106.5	105.7	+0.8%
e	$\chi_0 \to \chi_2$	$\lvert L_2\rvert^4$	$1/\sqrt{d_1}$	K_2	0.514	0.511	+0.7%
b	$\chi_0 \to \chi_2 \to \chi_3$	$\lvert L_2\rvert^4 L_3$	$d_2/(\sqrt{d_1}\,\lvert\zeta\rvert)$	K_0	4219	4180	+0.9%
s	$\chi_0 \to \chi_2 \to \chi_3$	$\lvert L_2\rvert^4 L_3^2$	$1/\lvert\zeta\rvert^2$	K_1	92.90	93.40	−0.5%
d	$\chi_0 \to \chi_2 \to \chi_3$	$\lvert L_2\rvert^4 L_3^2$	$1/\lvert\zeta\rvert^2$	$K_1\lambda^2$	4.58	4.67	−1.9%
c	$\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	K_0	1273	1270	+0.2%
t	$\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	K_0/α_{EM}	174.5 GeV	172.8 GeV	+1.0%
u	$\chi_0 \to \chi_2 \to \chi_3 \to \chi_1$	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	$K_0\lambda^2\alpha_W$	2.119	2.160	−1.9%
ν_3	$\chi_0 \to \chi_2 \to [\chi_0] \to \chi_2$	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	λ_3	50.67 meV	∼50	+1.3%
ν_2	$\chi_0 \to \chi_2 \to [\chi_0] \to \chi_2$	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	λ_2	10.01 meV	—	—
ν_1	$\chi_0 \to \chi_2 \to [\chi_0] \to \chi_2$	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	λ_1	5.113 meV	—	—

All 9 charged-fermion deviations are $\leq 2\%$; neutrino $\Sigma m_\nu = 65.79\,\text{meV}$, $R = 34.31$. All results derived from 4 L-function values $+N = \varphi(p^4)$; v, α_{EM}, α_W are experimental inputs.

7.2 Testable Predictions

(1) Neutrinos. This paper gives independent absolute-mass predictions for the three neutrino generations:

$$m_1 = 5.113\,\text{meV}, \qquad m_2 = 10.01\,\text{meV}, \qquad m_3 = 50.67\,\text{meV}.$$

m_1 and m_2 have not yet been directly measured; their detection would constitute a direct verification. Additional neutrino predictions:

- $\Sigma m_\nu = 65.79\,\text{meV}$: directly testable with CMB-S4 [24] sensitivity $\sim 30\,\text{meV}$;
- $R = \Delta m^2_{31}/\Delta m^2_{21} = 34.31$: independent of the phase δ and the absolute scale, this is the most central numerical prediction of this paper. JUNO [13] will measure it to $< 1\%$ precision; current $R_{\text{exp}} = 34.0 \pm 1.4$ [9] (0.2σ);
- $\delta_{CP} = -90°$: locked by the Gauss sum $\tau(\chi_3) = i\sqrt{7}$, not a fit. DUNE [28] and Hyper-Kamiokande [14] will measure it to $\pm 5°$;
- Normal ordering (NH): $m_3 > m_2 > m_1$. If JUNO confirms inverted ordering, the framework is falsified;
- Majorana type: $\chi_3 = \bar{\chi}_3$ (self-conjugate) requires neutrinos to be Majorana particles. Testable with $0\nu\beta\beta$ experiments (nEXO/LEGEND).

(2) Exact equalities between mass ratios and coupling constants. The three generations of up-type quarks share identical L-factors and node factors; mass ratios are entirely determined by generation factors. This yields two exact equalities:

$$\frac{m_t}{m_c} = \frac{1}{\alpha_{\text{EM}}(0)} = 137.036, \tag{20}$$

$$\frac{m_u}{m_c} = \lambda^2\,\alpha_W. \tag{21}$$

Eq. (20) binds the top/charm mass ratio to the electromagnetic coupling constant: current $m_t/m_c = 172800/1270 = 136.1$, deviation 0.7%. The HL-LHC and future e^+e^- colliders will advance the precision of top and charm masses to the 0.1% level; if m_t/m_c converges to 137.036 this constitutes strong confirmation; a deviation would falsify the framework. Eq. (21) similarly binds the u/charm ratio to the Cabibbo angle and the weak coupling.

(3) Higgs mass. Each of the three disjoint doublets D_1, D_2, D_3 of $(\mathbb{Z}/7\mathbb{Z})^*$ is split by the involution ι into a symmetric branch ($\iota = +1$, scalar) and an antisymmetric branch ($\iota = -1$, fermion), each with probability 1/2. The Higgs field, being a scalar, couples only to the symmetric branch within each doublet. The three doublets are independent; the joint probability of the scalar branches is:

$$\lambda_H = \left(\frac{1}{2}\right)^3 = \frac{1}{8}, \qquad m_H = v\sqrt{2\lambda_H} = \frac{v}{2} = 123.1\,\text{GeV}.$$

Experimental value: 125.10 GeV, deviation −1.6%.

(4) Structural predictions.

- **Exactly three generations**: arising from the rigid $\mathbb{Z}/3\mathbb{Z}$ structure. If a fourth-generation fundamental fermion is discovered, the framework is falsified.

7.3 Open Problems

Although this paper has connected most of the threads, several key open problems must be honestly stated:

- **RH/critical line problem**: This paper repeatedly exploits the phase-locking on the critical line and the rigidity of finite characters, suggesting that a Riemann-type spectral structure may participate at a deeper level, but the complete connection remains to be established.
- **Origin of $p = 7$**: Why does nature select precisely 7, rather than some other prime? This paper takes 7 as the minimal viable input, but its ultimate origin may require a higher-level selection principle or an absolute-geometry explanation [33, 34].

7.4 Conclusion

The 12 Standard Model fermion masses—from the top quark at 173 GeV to neutrinos at ~ 0.05 eV, spanning 12 orders of magnitude—were previously regarded as independent experimental inputs. This paper presents a unified formula:

$$\boxed{m_f = v \times |L|^{\text{pow}} \times \text{node factor} \times \text{Gen},} \tag{22}$$

reducing the 12 masses to the values of 4 Dirichlet L-functions on the critical line $s = 1/2$. All 9 charged-fermion deviations are $\leq 2\%$; the three neutrino absolute masses ($m_1 = 5.113$, $m_2 = 10.01$, $m_3 = 50.67$ meV, $\Sigma m_\nu = 65.79$ meV) and mass ratio $R = 34.31$ are derived within the same framework. The exact equalities $m_t/m_c = 1/\alpha_{\text{EM}}$ and $m_u/m_c = \lambda^2\alpha_W$ further bind the mass spectrum to the force coupling constants.

The $< 1\%$ measurement of R by JUNO (2026–2027), the ~ 30 meV sensitivity of CMB-S4 to Σm_ν, and the precise measurements of m_t/m_c and the Higgs self-coupling λ by HL-LHC will constitute direct tests of this framework.

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